IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

SELECTED TOPICS IN POWER SYSTEMS

Friday 12 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : B. Chaudhuri, F. Teng

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Answer ALL Questions

Question 1 on Flexible AC Transmission System (FACTS)

A transmission system operator plans to increase the transient stability limit (i.e. critical clearing angle/time) of an AC transmission line having a reactance of 0.4 pu. The system operator is considering either of the three line compensation options mentioned below under part (b), (c), (d). The voltages at the two ends of the transmission line are maintained at 1.0 pu and the equivalent system inertia is 4.0 sec. Calculate the critical clearing angle (in degrees) for each of the following cases considering 1.75 pu active power transfer. Assume a self-clearing fault and zero power transfer between the two ends of the line during the fault. Neglect the resistance and shunt susceptance of the transmission line.

a) Base case without any compensation.

[3]

b) 25% series compensation provided by a thyristor-controlled series capacitor (TCSC).

[3]

c) Ideal mid-point shunt compensation with a Static VAr Compensator (SVC) having a rated susceptance of 3.2 pu.

Make use of the formulae provided below:

Reactive power exchange with the SVC

$$Q_C = \frac{4}{X}\left(1 - \cos\frac{\delta}{2}\right)$$

Power-angle relationship once susceptance limit (B_{max}) of the SVC is reached

$$P = \frac{4}{X(4 - XB_{max})}\sin\delta$$

[8]

d) Ideal phase angle compensation with a Thyristor Controlled Phase Angle Regulator (TCPAR) having a steady-state (initial) phase angle compensation of $-5°$ and a compensation range of $\pm 22°$.

[6]

Question 2 on System Operation with Low Inertia

a) To maintain frequency stability in the future low-inertia power system, it is important to meet the system frequency requirements after large disturbances. The system dynamic is complicated but can be approximately described by the following swing equation:

$$2H\dot{f}(t) = R(t) - \Delta P_L$$

i) Draw a typical post-disturbance frequency evolution curve and identify the three frequency constraints that need to be maintained according to the regulation in GB.

[2]

ii) Analytically derive the rate of change of frequency (RoCoF), frequency nadir (at time instant t_n) and steady-state frequency constraints in terms of ΔP_L (largest loss MW), H (system inertia in [s]), R (frequency response [MW] linearly delivered by T_d), $RoCoF_{max}$ (maximum allowable rate of change of frequency), $\Delta f_{max\text{-}transit}$ (maximum allowable transient frequency deviation), and $\Delta f_{max\text{-}ss}$ (maximum allowable steady-state frequency deviation) from the swing equation, where you can neglect system damping and assume $t_n < T_d$.

[4]

iii) Considering two frequency services: R (Frequency Response that is linearly delivered by time T_d) and EFR (Enhanced Frequency Response that is linearly delivered by time T_{EFR}), the constraint that guarantees system frequency nadir within a maximum admissible frequency deviation is given below:

$$\left(H - \text{EFR}\times\frac{T_{EFR}}{4\times\Delta f_{max}}\right)\times R \geq \frac{(P_L - \text{EFR})^2 \times T_d}{4\times\Delta f_{max}}$$

Nomenclature

H: System level of inertia (MWs/Hz)
R: Amount of frequency response scheduled (MW)
EFR: Amount of EFR scheduled (MW)
P_L: Value of generation outage we want to secure against (MW)
Δf_{max}: Maximum admissible frequency deviation to avoid load shedding
T_d: Delivery time of frequency response (s)
T_{EFR}: Delivery time of EFR (s)

Other assumptions:

1. The largest generation unit is 1.8 GW
2. Other numerical values: H = 3 GWs/Hz, Δf_{max} = 0.8 Hz, T_d = 9.6 s, T_{EFR} = 0.64 s

The system operator has to procure the optimal amount of EFR and R to maintain the nadir constraint. In this operational time slot, the biding price of R is 62.25% of the biding price of EFR. Calculate the amount of EFR and R that should been procured to minise the total procument cost .

[4]

b) Answer the following questions related to the implementation of frequency support services in renewable generation systems:

i) Consider a wind turbine operating under Maximum Power Point Tracking at a medium wind speed (eg P=0.8 pu). Explain how the turbine would boost its output power by +0.1 pu to support the frequency regulation of the grid. Explain what limitations it would have (eg how long it would be able to boost the output).

[4]

ii) Calculate the equivalent synthetic inertia constant H if we are told that a power electronic converter increases its power output by 20% when the RoCoF of the grid is of -1.5 Hz/s in a 50 Hz grid.

[3]

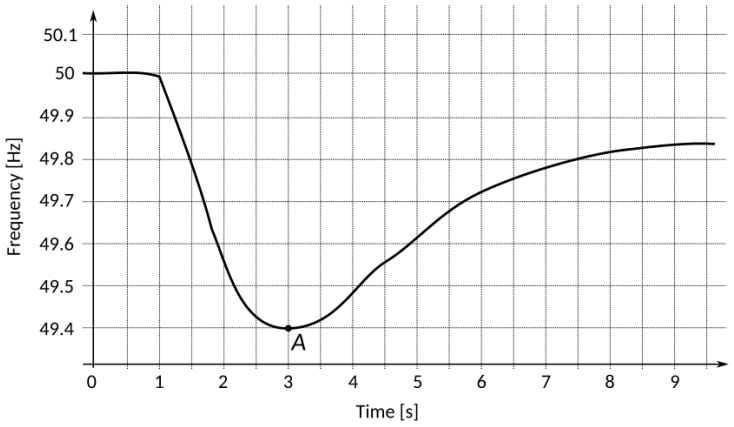


Figure 2.1 – Frequency excursion for Question 2.b.ii

iii) Consider a renewable generation system of 1 MW rated power that provides synthetic inertia with H=4 s and frequency droop with K=0.1 MW/Hz in a 50 Hz AC grid. The power setpoint of the generator was of 0.5 MW before the frequency excursion showin in Figure 2.1 started. Estimate what the power output of the generator will be once it reaches **point A** marked in the plot.

[3]

Question 3 on Power System State Estimation

(i) Why do system operators need state estimator?

[4]

(ii) What are the fundamental differences between state estimation in transmission and distribution network operation?

[4]

(iii) In the context of power transmission state estimation starting with measurement and states, formulate weighted least square (WLS) as maximum likelihood estimate (MLE).

[5]

(iv) What is equality constraints state estimation?

[3]

(v) Under what circumstances constraints state estimation is adopted?

[2]

(vi) What is the role of critical measurement in state estimation?

[2]

[20 marks]

Question 4 on Integrated Energy Systems

In the context of electrification of heat sector, heat appliances through thermal energy storage (TES) could provide flexibility to the electricity system and support more cost-effective integration of variable wind generation. Consider a system with a capacity of wind generation of 35GW, a nuclear generation of 13GW, and the rest of the load being supplied by CCGT plants, which are all the same type with the same characteristics. Electricity demand includes non-heat demand and electrified heat demand. Note that heat demand can be supplied by electricity and/or TES, therefore, TES can provide flexibility to the electricity system through shifting electrified heat demand. Operating reserve is required to deal with uncertainties in wind generation production, which is managed by part-loading CCGT.

Key characteristics of the CCGT plant are:

Capacity: 500 MW

Minimum Stable Generation (MSG): 300 MW

The generation cost function for each CCGT is illustrated below.

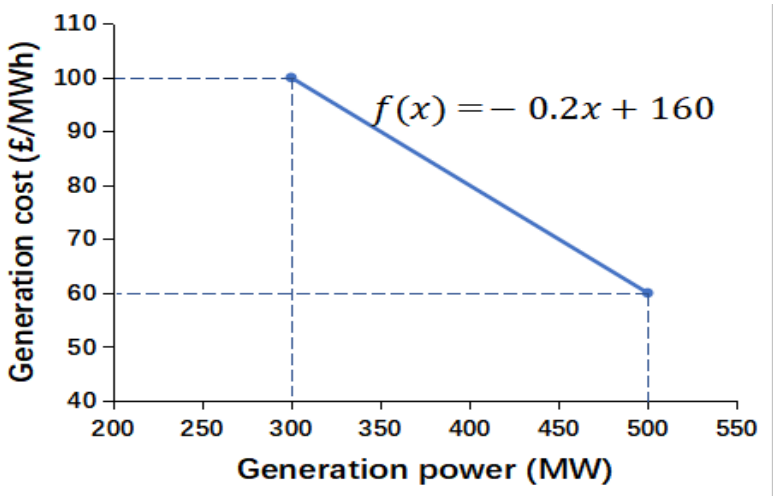


Figure 4.1 – Generation Cost Curve

a) Assume that TES is not available.
System operational conditions and reserve requirements are given in the table below, considering 2 consecutive hours of system operation.

Table 4.1 – System Data for 4.a)

	1st hour	**2nd hour**
Wind available (MW)	13,000	25,000
Nuclear (MW)	13,000	13,000
Electricity Demand (MW)	40,000	40,000
Reserve requirement (MW)	6,000	80,00

Answer the following questions:

i) In the first hour, how many CCGTs should be online?

[4]

ii) Based on the result of i), considering the most economical operating schedule, calculate the total lowest operating cost in the 1st hour.

[4]

b) Assume that TES is available, heat demand can be supplied by both electricity and TES. Surplus wind power can be converted into heat by electrified heaters (rated power is unlimited) and stored in TES. System operational conditions and reserve requirements are given in the table below, considering 2 consecutive hours of system operation.

Table 4.2 – System Data for 4.b)

	1st hour	**2nd hour**
Wind available (MW)	13,000	25,000
Nuclear (MW)	13,000	13,000
Non-heat electricity demand (MW) Heat demand (electricity equivalent) (MW_{el})	35,000 5,000	35,000 5,000
Reserve requirement (MW)	6,000	80,00

The parameters of TES are as follows:
- Maximum charging/discharging rate of TES is 7500MW_{el}.
- Total energy capacity of TES is 7500MWh_{el} (maximum thermal energy can be stored).
- The initial thermal energy stored in TES is 5000MWh_{el}.
- Heat loss of TES is ignored.

Answer the following questions:

i) Compared with question a) where TES was not available, determine how much less wind is curtailed in the 2nd hour.

[8]

ii) Consider demand response: reserve can also be provided by reducing electrified heat demand, i.e., reducing electrified heat demand by using TES instead. This is equivalent to increasing CCGT output to provide reserve. Note that 1MW electrified heat demand can provide 1MW reserve only if there is 1MWh_{el} thermal energy in TES.
Determine the lowest wind curtailment in the 2nd hour.

[4]